

Photosynthesis and Chlorophyll Fluorescence: Principles, Regulation, and PAM Fluorometry

Antoine Allard, Ludovic Jami and Marco Polin

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1 Introduction

Photosynthesis is the fundamental biological process through which light energy is converted into chemical energy [1]. It sustains nearly all ecosystems on Earth by producing organic carbon and molecular oxygen. In oxygenic photosynthetic organisms – including higher plants, algae, and cyanobacteria – photosynthesis takes place within chloroplasts, specifically across the thylakoid membrane. The process is traditionally divided into two interconnected stages: the light-dependent reactions, which convert light energy into ATP and NADPH, and the Calvin-Benson cycle, which uses these products to assimilate carbon dioxide.

Because photosynthesis is highly dynamic and sensitive to environmental conditions, researchers have developed non-invasive methods to probe its efficiency *in vivo*. Among these methods, chlorophyll fluorescence has become one of the most powerful and widely used techniques. Variations in chlorophyll *a* fluorescence provide direct information about the physiological state of Photosystem II (PSII), the balance between photochemistry and photoprotection, and the response of the photosynthetic apparatus to environmental stress.

Pulse-Amplitude-Modulation (PAM) fluorometry enables the study of photosynthesis by allowing fluorescence signals to be measured independently of ambient illumination. PAM measurements are now routinely used in plant physiology, marine biology, ecology, and biotechnology to estimate photosynthetic performance, monitor stress, and characterize photoprotective mechanisms.

2 The photosynthetic apparatus

2.1 Photosystems and light harvesting

Photosynthesis relies on two large pigment protein complexes embedded in the thylakoid membrane: Photosystem II (PSII) and Photosystem I (PSI) [1, 2]. These systems operate in series to drive linear electron transport from water to NADP^+ (Figure 2).

Light is absorbed primarily by chlorophyll molecules located within Light Harvesting Complexes (LHCs). The absorbed energy is transferred between pigments through resonance energy transfer until it reaches a reaction center. At the reaction center, charge separation occurs and initiates electron transport.

PSII is responsible for the oxidation of water and the release of oxygen. Electrons extracted from water pass sequentially through plastoquinone (PQ),¹ the cytochrome *b₆f* complex, plastocyanin, PSI, ferredoxin, and finally NADP^+ , forming NADPH.

The movement of electrons is coupled to proton translocation across the thylakoid membrane, generating a proton motive force (pmf). ATP synthase then uses this electrochemical

¹There are two types of plastoquinones: the primary plastoquinone acceptors, “A”; and the secondary plastoquinone acceptors, “B”. The former are stationary and permanently bound to the reaction centres. The latter are mobile and leave the reaction centre upon oxidation. Linear Electron Transport (LET) coordinates electrons from water to NADPH. A critical bottleneck is the two-electron gate at the PSII acceptor side:

1. Q_A (stationary) receives one electron.
2. Q_A^- reduces Q_B (mobile) to Q_B^- .
3. A second light reaction reduces Q_A again, which reduces Q_B^- to Q_BH_2 .
4. Q_BH_2 is released into the PQ pool and replaced by oxidized PQ.

Proper functioning of this gate requires bicarbonate ions. Additionally, Cyclic Electron Transport (CET) around PSI helps balance the ATP/NADPH ratio and protects systems against photoinhibition.

gradient to synthesize ATP.

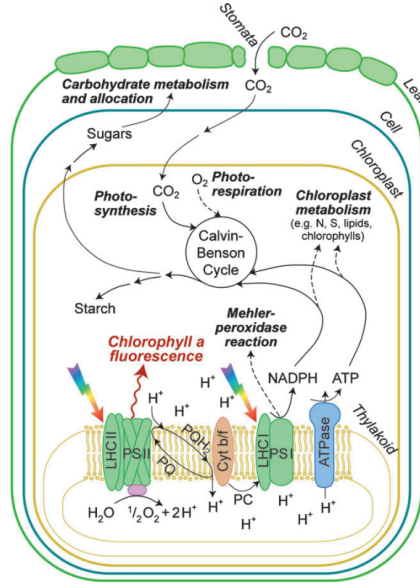


Figure 1: **Schematic representation showing the interactions of the main processes in C_3 photosynthesis in higher plants.** Photophosphorylation starts with the absorption of light by the light-harvesting antenna complexes (LHCI and LHCII) associated with photosystem II (PSII) and photosystem I (PSI) in the chloroplast thylakoid membrane. This drives electron transport from water via a series of electron carriers to NADP, producing reducing power (NADPH) and a H^+ electrochemical potential difference across the membrane. Dissipation of this proton motive force by the passage of H^+ back across the membrane through the ATPase drives the production of ATP. Ribulose 1,5-bisphosphate carboxylase/ oxygenase (Rubisco) catalyses the assimilation of CO_2 with ribulose 1,5-bisphosphate (RubP) in the carboxylation reaction of the Calvin–Benson cycle in the stroma of the chloroplast. The diffusion of atmospheric CO_2 into the leaf is regulated by the stomata in the epidermis. Other reactions of the Calvin–Benson cycle utilize NADPH and ATP to produce triose phosphates which are required for the synthesis of carbohydrates. NADPH and ATP are also used in a range of other metabolic activities (e.g. nitrogen and sulphur metabolism, lipid and pigment synthesis) in the chloroplast. Rubisco can also catalyse the photorespiratory oxygenation of RubP, which involves consumption of NADPH and ATP by the Calvin–Benson cycle. O_2 can also be directly photoreduced by electron transfer from PSI to produce superoxide which is then rapidly dismutated to hydrogen peroxide, which in turn is detoxified to water; this process is often termed the Mehler-peroxidase reaction or the water-water cycle. Cyt b/f , cytochrome b_6/f complex; PC, plastocyanin; PQ, plastoquinone; PQH2, plastoquinol. From [3].

2.2 Energy pathways

Once a chlorophyll molecule absorbs a photon, the excited state must relax through one of three competing pathways [4]:

1. *photochemical quenching*: energy is used to drive electron transport, meaning that the energy is used for photosynthesis;
2. *non-photochemical quenching*: excess excitation energy is dissipated by the cell;
3. *fluorescence*: energy is re-emitted as light.

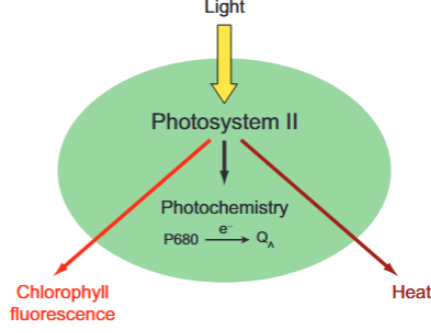


Figure 2: **Simple model of the possible fate of light energy absorbed by photosystem II (PSII).** Light energy absorbed by chlorophylls associated with PSII can be used to drive photochemistry in which an electron (e^-) is transferred from the reaction center chlorophyll, P680, to the primary quinone acceptor of PSII, Q_A . Alternatively, absorbed light energy can be lost from PSII as chlorophyll fluorescence or heat. The processes of photochemistry, chlorophyll fluorescence, and heat loss are in direct competition for excitation energy. If the rate of one process increases the rates of the other two will decrease. From [5].

The fluorescence yield of PSII is strongly coupled to the redox state of the primary quinone electron acceptor, with $[Q_A]$ the fraction of oxidised plastoquinones “A” in the population.

- When Q_A is oxidized, reaction centers are considered *open* and can accept electrons efficiently. Photochemistry proceeds rapidly, resulting in relatively low fluorescence.
- When Q_A becomes reduced (Q_A^-), reaction centers become *closed*, preventing further charge separation. Photochemical quenching decreases and fluorescence rises.

These three routes of energy usage/dissipation are characterised by the following rates:

- $k_P[Q_A]$ is the rate for *photochemical quenching* (dark adapted cells);
- k_{NPQ} is the rate constant for *non-photochemical quenching* (dark adapted cells);
- k_F is the rate constant for *chlorophyll autofluorescence*,

where these refer to the rates when cells are **dark adapted**.

The total rate of energy usage is the sum of the three rates, and the relative probability of one process over the others is the ratio between the rate of the process and the sum of the rates. For example, this means that, given the three rates above, the fluorescence signal would be proportional to the fluorescence yield:

$$\Phi_F = \frac{k_F}{k_F + k_{NPQ} + k_P[Q_A]} \quad (1)$$

2.3 Reaction centers and fluorescence yield

Since fluorescence is inversely related to photochemical activity (Eq. 1), chlorophyll fluorescence acts as an indirect probe of photosynthetic electron transport. Using chloroplast autofluorescence it is possible to estimate physiological parameters like PSII operating efficiency, and hence Linear Electron Flow (LEF).

2.3.1 Measurements for dark adapted cells

The energy usage rates above depends on the state of light adaptation of the cell. For dark adapted cells, let us define:²

- $F_0 \propto \frac{k_F}{k_F + k_{NPQ} + k_P}$: fluorescence of dark adapted cells ($[Q_A] = 1$).
- $F_M \propto \frac{k_F}{k_F + k_{NPQ}}$: *maximum* fluorescence of dark adapted cells ($[Q_A] = 0$).
- $F_v = F_M - F_0$: variable fluorescence.

When dark-adapted cells are kept in the dark, most PSII reaction centers are open ($[Q_A] = 1$), the fluorescence measured under weak measuring light is therefore minimal, and denoted F_0 .

If a saturating flash transiently closes all PSII reaction centers ($[Q_A] = 0$) without affecting other parts of chloroplast physiology, photochemistry is no longer possible, and the rate of photochemical quenching is 0. The fluorescence increases to its theoretical maximum F_M .

If we take the ratio of the variable and the maximum fluorescence, the proportionality constants cancel out and we obtain:

$$\frac{F_v}{F_M} = \frac{k_P}{k_F + k_{NPQ} + k_P} = \Phi_{PSII}^{\max}. \quad (2)$$

known as the maximum photochemical quantum yield of PSII. It is commonly used to monitor plant stress, e.g. due to sub-optimal environmental conditions. Stress would lower $\Phi_{PSII}^{\max}$ below the level characteristic of that organism when grown optimally.

2.3.2 Measurements for light adapted cells

While the maximum quantum yield of PSII photochemistry is commonly achieved only at *very low* ambient light, under light conditions, PSII will operate with a different level of photochemical efficiency. This is due to cells adapting to environmental light conditions.

For light-adaptation we will define:

- k'_P : maximum rate constant for photochemical quenching (light conditions);
- k'_{NPQ} : rate constant for non-photochemical quenching (light conditions);

These quantities will depend on the light intensity (I) and on time (adaptation is not immediate). The fluorescence rate k_F is assumed to be independent of light conditions.³ The rates in light conditions can be used to define [6]:

- $F' \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P [Q_A]}$: operating fluorescence in light conditions;
- $F'_0 \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P}$: minimum fluorescence in light conditions;
- $F'_M \propto \frac{k_F}{k_F + k'_{NPQ}}$: maximum fluorescence in light conditions;

²Note that the proportionality constant will depend on the concentration of cells and on the details of the experimental apparatus, and it is not generally possible to measure it accurately. The good thing is that, if we combine measurements, it is not necessary to know it in order to probe chloroplast activity. It is important to remember that often we are interested in changes in fluorescence parameters (e.g. how do they correlate with other behavioural traits) rather than estimating their absolute values.

³High light can cause direct photodamage of the absorbing molecules.

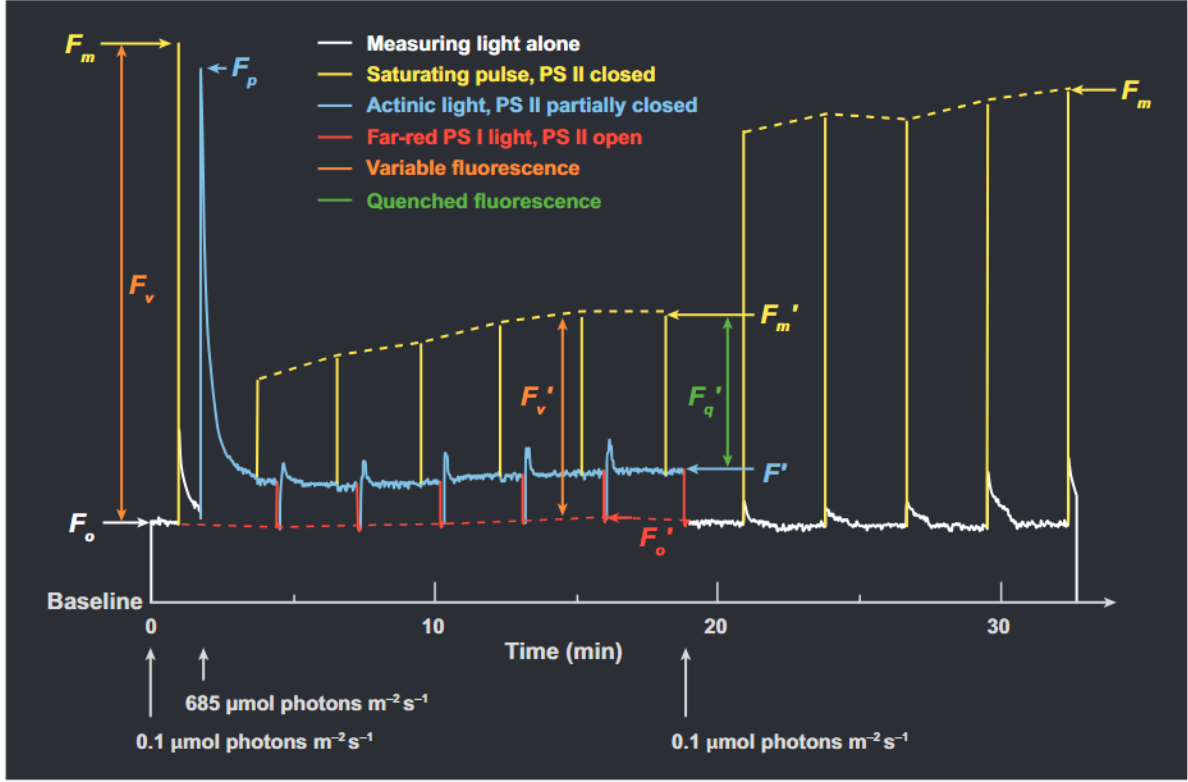


Figure 3: Protocol for quenching analysis using modulated fluorescence. A dark-adapted leaf is exposed to various light treatments. The fluorescence parameters denoted with a prime originate from the illuminated leaf, where energy-dependent, non-photochemical quenching is present. The parameters without a prime are obtained from the leaf in the dark-adapted state, where there is no energy-dependent non-photochemical quenching. The different colours of the trace denote different light treatments. White, weak measuring light alone ($0.1 \mu\text{E}$) that gives F_0 . Yellow, saturating light pulse ($<1 \text{ s}$ duration, $>6000 \mu\text{E}$) that gives F_m in darkness and F'_m in light. Blue, actinic light that drives photosynthesis (in this case $685 \mu\text{E}$) that gives F' (if steady-state has been reached this has often been denoted by F'_s). The actinic light can be produced from a range of sources, for example, sunlight, halogen lamp, light-emitting diodes. The initial peak of fluorescence is denoted as F_p (without prime, since it originates from the nomenclature of the rapid phase of fluorescence induction). Red, far-red light ($30 \mu\text{E}$ at $720\text{--}730 \text{ nm}$ for 4 s) that excites PSI only, and thus oxidizes the plastoquinone and Q_A pools associated with PSII and gives F'_0 : Orange, variable fluorescence calculated as $F_v = F_m - F_0$ from the dark-adapted leaf and $F'_v = F'_m - F'_0$ from the illuminated leaf. Green, fluorescence that is quenched from F'_m to F' by PSII photochemistry in the illuminated leaf, calculated as $F'_q = F'_m - F'$. All parameters, except F'_q , F_v , and $F' - v$; are measured from the baseline. From [3].

- $F'_v = F'_m - F'_0 = \frac{k_F k'_P}{(k_F + k'_{NPQ})(k_F + k'_{NPQ} + k'_P)}$: variable fluorescence in light conditions;
- $F'_q = F'_m - F' = \frac{k_F k'_P [Q_A]}{(k_F + k'_{NPQ})(k_F + k'_{NPQ} + k'_P [Q_A])}$: quenched fluorescence in light conditions.

From these we can define:

- $\frac{F'_v}{F'_m} = \frac{k'_P}{k_F + k'_{NPQ} + k'_P} = \Phi_{\text{PSII}}^{\text{max}}(I)$, maximum PSII photochemical efficiency at the given light level;

- $\frac{F'_q}{F'_v} = [Q_A] \frac{k_F + k'_{\text{NPQ}} + k'_P}{k_F + k'_{\text{NPQ}} + k'_P [Q_A]}$, PSII efficiency factor. This is the fraction of the maximum PSII efficiency at the given light level, that is actually realised. It is determined by the ability of the photochemical apparatus to keep the plastoquinones oxidised;
- $\frac{F'_q}{F'_M} = \frac{k'_P [Q_A]}{k_F + k'_{\text{NPQ}} + k'_P [Q_A]} = \frac{F'_q}{F'_v} \times \frac{F'_v}{F'_M} = \Phi_{\text{PSII}}(I)$, PSII operating efficiency at the given light level.

Determining F'_v/F'_M and F'_q/F'_v allows one to assess whether changes in PSII operating efficiency are due to changes in NPQ or on the ability of PSII to drive electron transport.

The following identities can also be useful:

- If we assume that $k_P = k'_P$, F'_0 can be calculated from F_0, F_M, F'_M as $F'_0 = \frac{F_0}{(F_v/F_m) + (F_0/F'_m)}$
- $[Q_A]$ can be estimated as $[Q_A] = \left[\frac{F'_q}{F'_v} \right] \left[\frac{F'_0}{F'_v} \right]$
- NPQ can be estimated also as $\frac{F_M - F'_M}{F'_M} = \frac{k'_{\text{NPQ}} - k_{\text{NPQ}}}{k_{\text{NPQ}}}$ (see this paper and reference [21] within) CITE

3 Factors affecting PSII efficiency

There are two main types of factors affecting PSII efficiency: photochemical quenching and non-photochemical quenching.

Photochemical quenching. This refers to changes in efficiency due to the photochemical processes of the chloroplast, which ultimately lead to the formation of ATP and NADPH. The main factor influencing PSII efficiency is $[Q_A]$, which can be monitored as indicated above.

Non-photochemical quenching (NPQ). NPQ represents a suite of photoprotective mechanisms that prevent photo-oxidative damage. Excess excitation energy can generate triplet chlorophyll states, which react with molecular oxygen to produce singlet oxygen and reactive oxygen species (ROS). NPQ reduces this risk by safely dissipating excess energy as heat.

When the chloroplast absorbs light energy at a rate that is too high compared with what can be utilised through photochemical quenching or emitted by fluorescence, there is an accumulation of excited electrons in the light harvesting complexes. This is a problem for the cell. For example, O_2 can absorb the energy of excited electrons and split into singlet oxygens, which are extremely reactive. Photochemical quenching might also need to be reduced to control production of ATP and NADPH. In these cases, the cell can activate a suite of biochemical processes (NPQ) to protect the photosynthetic apparatus and guarantee that:

- i) the energy captured is not excessive. The rate of energy absorption should be matched to the level that the system can effectively handle;
- ii) the energy captured is well balanced between PSI and PSII. PSI is involved in both linear and cyclic electron flows (LEF, CEF), while PSII takes part only in LEF.

There are three contributions to NPQ:

- q_E energy-dependent quenching: the fastest and most important component of NPQ. q_E is involved in regulation of energy capture (point i) above) Strong illumination increases proton accumulation within the thylakoid lumen, decreasing luminal pH. Acidification

activates the xanthophyll cycle: Violaxanthin $\rightarrow$ Antheraxanthin $\rightarrow$ Zeaxanthin. Protonation of zeaxanthin leads to a conformational change in PSII reaction centres and promotes thermal dissipation of excitation energy. In higher plants, the **PsbS protein** acts as a pH sensor and quenching amplifier. In algae like *Chlamydomonas*, this role is fulfilled by the **LHCSR3** protein.

- q_T state-transitions quenching: change in the way light-harvesting complexes (LHCs) are associated to PSI or PSII, and/or changes in whether LHCs can absorb light or not (open=absorbing light; closed= not absorbing light). q_T is involved in both regulation of energy capture and PSI/PSII balance (points i) and ii) above). This can involve up to 80% of the mobile LHCs in *Chlamydomonas* but only $\sim 20\%$ in land plants.
- q_I photoinhibitory quenching: the slowest, photodamage due to excessive light intensity. In high light, a high percentage of closed PSII centers causes a “backpressure” in the electron transport chain, which can lead to reversible inactivation or inhibition of photosynthesis, known as dynamic photoinhibition. It involves damage to the **D1 protein** of the PSII reaction center and requires disassembly of the PSII complex and synthesis of new D1 for repair.

The different processes happen over a range of timescales that typically vary between seconds (even faster) to minutes.

4 Pulse-Amplitude-Modulation (PAM) Fluorometry

4.1 Principles of PAM Fluorometry

Pulse-Amplitude-Modulation fluorometry was developed to separate chlorophyll fluorescence from ambient actinic illumination [7].⁴ Although this is an established technique, care must be taken when performing experiments and interpreting the data. The technique relies on three different light sources:

1. **Measuring Light (ML)**: very weak, modulated light used to measure fluorescence without significantly driving photosynthesis.
2. **Actinic Light (AL)**: continuous illumination driving photosynthesis and physiological adaptation.
3. **Saturating Pulse (SP)**: brief and extremely intense flash that temporarily closes all PSII reaction centers.

⁴Beyond PSII fluorescence, other signals provide a “multiscopic” view.

- **Electrochromic Shift (ECS)**: an absorbance shift (near 520 nm) reflecting the trans-thylakoid electric field ($\Delta\Psi$). It is used to probe pmf and ATP synthase conductivity.
- **P700 spectroscopy**: absorbance changes at 830 nm monitor the redox state of the PSI reaction center, allowing the estimation of PSI quantum yield (Φ_I).

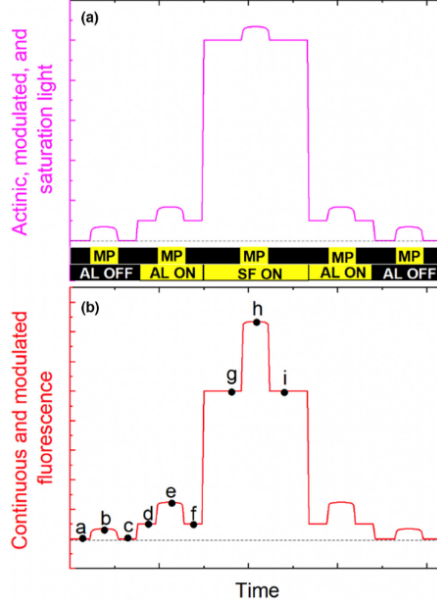


Figure 4: **Schematic depiction of light sources used during experiments** (a) cumulative light intensities that are produced by the modulated light (ML), actinic light (AL), and saturation flash (SF) light sources are schematically depicted. Individual modulated pulses (MPs) are shown at the bottom of the figure. The bar at the very bottom represents continuous, non-modulated light sources, AL and SF light; (b) the cumulative fluorescence signals elicited by the respective light sources are schematically depicted; modulated fluorescence (MF) and continuous fluorescence (CF) are elicited by MPs and AL and SF, respectively. The letters represent time points at which the ML and detector electronics facilitate the detection of fluorescence intensities. Conceptually speaking, the MF signal is electronically demodulated, or discriminated, from the CF signal according to the expression: $MF = e - (d + f)/2$ where MF is measured during all non-modulated illumination conditions based on the same electronic principles (see points a through c and g through i, respectively). The CF signal is depicted as being measured by detecting fluorescence intensities at points d, f, g, and i, and so on. From [8].

4.2 Dark-Adapted Measurements

Dark adaptation typically lasts 20 to 30 minutes and allows most Q_A molecules to return to the oxidized state.

The maximum quantum yield is defined as:

$$\frac{F_v}{F_M} = \frac{F_M - F_0}{F_M}$$

This parameter estimates the maximum efficiency with which absorbed light can drive PSII photochemistry.

A reduction in F_v/F_M usually reflects environmental stress such as excessive light, nutrient limitation, temperature stress, etc.

4.3 Light-Adapted Measurements

Under illumination, photosynthetic organisms dynamically regulate electron transport and energy dissipation.

In light-adapted conditions, the following parameters are defined:

- F' : steady-state fluorescence,
- F'_M : maximal fluorescence in the light,
- F'_0 : minimal fluorescence in the light.

The operating quantum yield of PSII is:

$$\Phi_{PSII} = \frac{F'_q}{F'_M} = \frac{F'_M - F'}{F'_M}$$

This parameter estimates the fraction of absorbed light used for photochemistry under a given light intensity. In addition, non-photochemical quenching is commonly estimated using:

$$NPQ = \frac{F_M - F'_M}{F'_M}$$

An increase in NPQ indicates enhanced thermal dissipation of excess excitation energy.

4.4 Summary of Measurement Protocols

1. **Dark Adaptation:** 20–30 min to ensure Q_A is oxidized.
2. **Determine F_0 and F_M :** Use ML and a SP.
3. **Actinic induction:** Apply light and monitor the fluorescence yield.
4. **Steady-state:** Measure F' and apply SP for F'_M to calculate Φ_{PSII} and NPQ.

5 Conclusion

Photosynthesis is a highly regulated process that balances efficient energy conversion with protection against photo-oxidative damage. Chlorophyll fluorescence provides a unique window into this balance because fluorescence, photochemistry, and heat dissipation are intrinsically linked.

Pulse-Amplitude-Modulation fluorometry allows accurate measurements of PSII efficiency under realistic conditions. Parameters such as F_v/F_M , Φ_{PSII} , and NPQ are now central tools for understanding photosynthetic performance and stress physiology. The integration of PAM fluorometry with fluorescence imaging, spectroscopy, and mathematical modeling has provided a multiscale view of photosynthetic regulation.

List of abbreviations

AL	Actinic Light
ATP	Adenosine Triphosphate
CET	Cyclic Electron Transport
CF	Continuous Fluorescence
Cyt <i>b/f</i>	Cytochrome <i>b₆/f</i> complex
ECS	Electrochromic Shift
LEF	Linear Electron Flow
LET	Linear Electron Transport
LHC	Light Harvesting Complex
MF	Measuring Fluorescence
ML	Measuring Light
NADPH	Nicotinamide Adenine Dinucleotide Phosphate oxidase
NPQ	Non-photochemical quenching
PAM	Pulse Amplitude Modulation
PC	Plastocyanin
pmf	proton motive force
PSI	PhotoSystem I
PSII	PhotoSystem II
PQ	Plastoquinone
PQH ₂	Plastoquinol
Q_A	Plastoquinone acceptor A
Q_B	Plastoquinone acceptor B
ROS	Reactive Oxygen Species
Rubisco	Ribulose 1,5-bisphosphate carboxylase/ oxygenase
RubP	Ribulose 1,5-bisphosphate
SF or SP	Saturating Flash or Pulse

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